From Twinhood to Selfhood: The Evolution of Digital Human Twins Through Affordance Learning

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Abstract

The rise of digital human twins (DHTs) signals a shift beyond industrial twins toward questions of cognition and autonomy. Traditionally treated as passive digital replicas, DHTs are poised to become agents that perceive, decide, and learn within increasingly data-rich virtual environments (VEs). This paper challenges the mirror metaphor: DHTs need not merely reflect their human counterparts but can develop affordance-based perceptions of virtual space. Building on ObjectOriented Ontology (OOO), we treat DHTs as non-human digital objects only partially accessible through their interfaces; extending affordance theory with Artifact–Artifact Affordance (AAA), we analyze interactions that are not centered on human perception. We propose a conceptual DHT Autonomy Continuum and a typology of agent–environment couplings, arguing for a synergy in which computational methods operationalize ecological (direct) coupling. Using inductive reasoning—drawing on reinforcement-learning principles and agent-based affordance acquisition—we theorize how DHTs could transition from passive replicas to partially autonomous digital agents without exhaustive human programming. The contribution is theoretical: examples function as illustrations of possibility rather than evidence of deployed systems, and the aim is to inform design, evaluation, and governance of DHT–VE ecosystems.

Keywords: human digital twins, digital human twins, virtual environments, AI autonomy, Object-Oriented Ontology, Artifact-to-Artifact Affordance, agent-environment coupling, ecological psychology, computational psychology, reinforcement learning, affordance learning, autonomy continuum

1 Introduction

1.1 The Digital-Physical Duality of Human Twins

With rapid advancements in digital twin technologies, including the Internet of Things (IoT), 5G/6G, and artificial intelligence (AI), the concept of digital twins has expanded significantly. Traditional digital twins focus mainly on industrial and physical systems, where data parameters are exchanged between the physical and digital realms. In this

context, digital shadows serve as automated reflections of real-world changes, while digital twins go further by enabling bidirectional updates, allowing for continuous synchronization and optimization of physical assets in digital environments (Sulaiman Khail and Schnizer, 2024).

Digital Human Twins (DHTs), as an emerging concept, has extended the digital twin applications to human anatomy and cognition. DHTs by definition are virtual representations of human beings that dynamically mirror and model their physical counterparts. Unlike conventional digital twins, DHTs encompass more dynamic human attributes, including physiological, cognitive, and decision-making processes (Gaffinet et al, 2025a; Miller and Spatz, 2022; Lin et al, 2022). This fundamental difference introduces additional layers of complexity, as updates in DHTs involve not only real-time data but also the integration of intangible aspects of human.

In current engineering practice, the applications of DHTs are useful in various fields. In personalized healthcare, a DHT could simulate responses to medication or treatment plans, helping doctors make more informed decisions (Gaffinet et al, 2025b). For instance, a patient's twin could analyze real-time data from wearable devices to predict the onset of illnesses before symptoms manifest, offering proactive health interventions. In workplace ergonomics, a DHT can help employers to monitor the physical strain of employees during tasks, suggesting adjustments to reduce fatigue and prevent injuries (Lin et al, 2022). In military applications, DHTs can simulate human behavior in complex operational environments to improve training and decision-making. For example, a DHT might simulate a soldier's cognitive and physical performance under stress, helping to design better equipment and training programs (Miller and Spatz, 2022). Similarly, in disaster management, DHTs can assist in optimizing humansystem interactions during emergency responses. In education, DHTs are poised to transform personalized learning. A twin could analyze a student's strengths and weaknesses, adaptively recommending study plans or identifying knowledge gaps (Lin et al, 2022). This capability is particularly valuable for remote learning environments where tailored support can enhance engagement and outcomes.

While much recent work on DHTs has focused on technical feasibility—data fusion, AI integration, and clound computing—there remains a lack of conceptual clarity about what a DHT is as an ontological entity. Shengli (2021) notes that most DHT research addresses narrow use cases and overlooks the full lifecycle of human beings, leaving unresolved challenges not only of complexity and modeling but also of ethics and legitimacy. This paper responds to that gap by developing a philosophical and conceptual account of DHTs, which we treat as virtual agents with evolving sensorimotor contingencies in virtual environments. In addition to the duality of DHTs, our conceptual DHT model elaborates the environmental aspects that Shengli (2021) emphasizes to make DHT model more comprehensive.

1.2 Virtual Environments as Perception-Action Testing Grounds

For human, the physical environments provide information cues, offering them possibilities for actions. Same applies to artificial environments:

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> It is a mistake to separate the natural from the artificial as if there were two environments ... There is only one world, however diverse, and all animals live in it, although we human animals have altered it to suit ourselves (Gibson, 1986, p. 130).

Virtual environments (VEs), the artificial worlds we human create for artificial beings (i.e. DHTs), can be used as the primary testing ground for controlled laboratories studying systematically how DHTs perceive and interact with their environments. One might argue that VEs are not real and do not provide real experiences as the physical worlds for humans. However, Chalmers (2017) from a realist stance argues that VEs are a form of genuine reality, and virtual objects as non-illusory digital entities are constituted by computational processes and data structures. Human experiences involving virtual objects within VEs are considered non-illusory perceptions of the digital world, thus virtual objects' reality is grounded in VEs, much like physical objects are grounded in the physical world. In what follows, we use thisas a conceptual assumption to grant talk of DHTs engaging with VE objects as "real" for them, independently of whether any particular system implements this today. Agreeing with Harman (2018) on that non-human entities possess an ontological reality independent of human perception, DHTs' reality is grounded in the VEs where they perceive and interact with virtual objects– they are genuinely engaging with entities that exist within VEs. In the VEs testing grounds, unlike in physical reality where natural laws govern, DHTs potentially can experience new, non-human modes of interaction with the VEs which can be enriched with dynamic and multidimensional data
(i.e. audio-visual, tactile, kinetic data).

1.3 Research Objectives

Current research predominantly treats DHTs as passive representations of their human counterparts, designed primarily for mirroring behavior and decision-support tasks. However, drawing on Object-Oriented Ontology (OOO), Umwelt theory, and ecological psychology, we challenge the mirror metaphor and propose that DHTs are not mere digital replicas, but evolving autonomous agents capable of developing their own perception-action models in VEs. Through a multi-disciplinary approach, we theorize DHTs autonomy and non-human agency. The position we develop in this paper contributes to the development of the conceptual framework of DHTs Autonomy Continuum and the agent-environment coupling typology.

This paper is organized into the following sections. In Section 2, we introduce OOO as a primary philosophical framework and compare computational psychology and ecological psychology when addressing the agent-environment coupling, following by bridging affordance theory with non-human theory. In Section 3, we then examine the twin vs. autonomy paradox in DHTs, and propose the DHTs Autonomy Continuum as a conceptual solution to this paradox. The DHTs Autonomy Continuum framework defines five levels of independence in digital twins. Here we argue that OOO explains *what* a DHT is ontologically (an entity with its own agency), while ecological psychology explains *how* a DHT perceives and learns (through affordances and interaction). Building on this foundation, we propose a computational ecological view to the evolution of DHTs

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autonomy, emphasizing the synergy of both in shaping affordance perception, as illustrated through the example of walking on stairs in VEs. Section 4 discusses the ontological shift that integrates OOO and AAA, and argue that a synergistic ecological–computational coupling—framed by affordances and AAA—explains how they can exceed mere simulation through learning in VEs. The discussion extends to a broader discourse on the evolving identity of digital twins and ethical concerns. Finally we conclude the contributions of this paper and propose future research directions.

Examples in this paper are illustrative, not demonstrations of existing systems. We adopt, by assumption, a realist stance toward VEs—virtual objects are treated as real-for-the-agent to license talk of non-human couplings—without claiming presentday implementations. Claims are VE-bounded; transfer to physical environments and empirical thresholds between levels remain open questions, and we do not advance a specific normative theory of rights.

2 Background and Related Work

2.1 Object Oriented Ontology (OOO)

Object-oriented ontology (OOO) is a metaphysical stance within the broader speculative-realist turn that refuses to make human access the measure of *what is* or *being*. As framed by Graham Harman by deriving the term from the object oriented programming languages (Harman, 2011a; Bryant, 2011), OOO holds that objects—natural and artificial, living and non-living, digital and physical—exist without being reduced to their relations or to their access to human cognition. In this respect, OOO urges a "flat" ontology denying any principled ontological hierarchy that would privilege human subjects over the rest of reality.

Harman's key move is to distinguish between *real* objects and *sensual* objects (Harman, 2011b, 2018). Real objects withdraw: they retain a surplus of being that no encounter, description, or use can fully disclose. Sensual objects are the ways entities appear in experience and relation. Because withdrawn entities never touch directly, causation itself is vicarious—objects interact only through the mediated profiles by which they enter into contact (Harman, 2011a). This is why any attempt to reduce an object to its use or appearances misses the point. A hammer, for instance, is not reducible to the way it looks in front of us or to its use for pounding nails. It can also decay, splinter, be melted down, or serve as a paperweight. Each relation reveals only a slice of its being.

Building on OOO's insight, we can turn to Von Uexku¨ll (1920, 2013)'s work on *Umwelt*. Every organism inhabits a world structured by its own capacities to sense and act; its *Merkwelt*, the field of salient cues, filters an inexhaustible reality into a workable environment. Famously, the tick's Umwelt is reduced to temperature, odor, and touch, producing a radically different lived reality than that of a human. Set besides OOO, Uexku¨ll shows that perspective gives a partial world, while OOO insists that reality is bigger than any part we see. Just as an organism's Merkwelt discloses only a fraction of its Umwelt, any relation discloses only a profile of a withdrawn object. The two lines of thought agree on one thing: reality is bigger than what we can access.

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This convergence matters for digital artifacts and agents. Contemporary practice often flattens digital objects to code, outputs, or user interfaces (UI). OOO resists that flattening—a DHT, a replica of human is not exhausted by its appearance, Application Programming Interface (API) surface, logs, or UI. Withdrawal keeps us aware that hidden potentials, latent structures, and failure modes count for how such systems couple to their VEs and other entities. In Harman's terms, what contacts other entities are sensual profiles—Graphical User Interfaces (GUIs), APIs, data schemas, embeddings—through which vicarious causation operates; what remains withdrawn are internal states, training histories, emergent learning, and constraints that never fully appear.

From Uexku¨ll's standpoint, an AI-empowered DHT in VEs inhabit an Umwelt determined by their sensors, effectors, reward signals, and data structures: what they can register (their Merkwelt) and what they can do shape the "world" they experience. Alter the environment's parameters (affordances, latency, topology, feedback) and you reconfigure the agent's Umwelt, changing salience and interaction dynamics. From an OOO perspective, those interactions reveal only partial profiles of both parties, and the DHT never fully grasps the VE, and the VE never fully "shows" itself to the DHT. This framing also aligns with Actor-Network Theory (ANT)'s emphasis on heterogeneous, non-hierarchical networks of actants (Latour, 2005), but OOO adds the ontological reminder that every actant (i.e., human, nonhuman, digital being) withdraws beyond its roles in the network.

The OOO serves a preliminary philosophical step in conceptualizing DHT–VE couplings as genuine ontological interactions, not metaphors or anthropomorphic extensions; we now turn to the problem of agent–environment coupling to understand how, in principle, a DHT perceives, acts, and learns in a VE.

2.2 On the Fundamental Problem of Agent-Environment Coupling

How an agent couples to its environment is not a side issue—it is the condition of possibility for perception, action, learning, and control. *Coupling* indicates the ongoing, bidirectional relation in which an agent's actions reshape the very inputs it later receives, while the environment's structure constrains and enables what the agent can do next. Debates over how agents relate to their environments have often polarized into two positions: an indirect, symbol-mediated view associated with Herbert A. Simon, and a direct, ecological view associated with James J. Gibson. Shaw (2003), in "The Agent–Environment Interface: Simon's Indirect or Gibson's Direct Coupling?", sets this contrast explicitly and uses it to probe what genuine coupling requires.

Simon (1969, 2019) treats agent-environment coupling as *artifactual* and *rule-governed*, meaning interaction is mediated by symbol realized through information processing. His simulation thesis suggests:

> Instead of just studying experimentally and observing naturalistically how humans adapt to a complex environment, one might instead and more conveniently study computer simulations of such adaptive behaviors, with realistic expectations of discovering the design principles underwriting their organizations (Simon, 2019, p. 138).

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This line underwrites many current practices with DTs and DHTs, for example Siemens's digital-human simulations for planning and control (Siemens, 2023); and patient-specific cardiac and brain digital twins (e.g., the Living Heart Project (Baillargeon et al, 2014), EPINOV (Syst`emes, 2020) that forecast treatment response and guide clinical decisions.

Gibson (1979) offers a contrasting proposal: the interface is *natural* and *law-governed*, a *direct* informational coupling rather than a symbolically mediated one. On this view, perception is not a reconstruction from inner codes but an attunement to information available in ambient optic array (No¨e, 2025). Shaw sharpens the contrast by arguing that open-ended context defeats any bid to pre-specify exhaustive rule sets:

> No rule for predicting the contextual exigencies can be written; no way is known for simulating specific contexts because their denotative richness and their connotative implications are unbounded... Because any radical simulation project inescapably encounters this paradox, we would do well to examine it (Shaw, 2003, p. 47).

Placing emphasis on direct perception, Shaw defends the ecological alternative that nature solves situated actions not by cataloging all possible rules in advance, but by equipping agents with perceptual systems that disclose task-relevant structure in *real time*. Unlike a programmer who must pre-define every possible condition in advance, an agent's awareness of environmental details grounds its capacity to act. This awareness enables the agent to engage effectively with its surroundings and realize its goals, without collapsing into an infinite regress of pre-specified rules. Together, these positions motivate a turn to affordance theory to analyze coupling in terms of situated possibilities for action that emerge for a given agent with given capacities in a given environment.

2.3 Concept of Affordance

Gibson coins the term *affordance* to designate the action possibilities the environment offers to a particular organism, given that organism's capacities; affordances are neither purely objective nor purely subjective, but relational—they "point both ways, to the environment and to the observer" (Gibson, 1977, 1979). On the ecological view, perception is an active process in which agents directly pick up these possibilities as they move within structured ambient optic array, rather than building behavior from internal symbols alone (also see Section 2.2) (Gibson, 1979). In the physical world, humans effortlessly integrate affordances into their cognitive and perceptual systems, allowing fluid interaction with their surroundings. This process occurs through sensorimotor learning, where experience refines an individual's ability to recognize and act upon affordances in their environment. A core aspect of Gibson's theory is the distinction between affordances that exist whether noticed or not and perceived affordances that depend directly on an agent's sensory–motor capacities. A chair affords sitting whether or not someone recognizes it; a child may fail to perceive that a door handle affords turning if the requisite skills are missing. Extending these ideas to HCI, Norman (1999) emphasizes making affordances perceptible and intuitive, while Gaver (1991) and

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Stoffregen (2000) underscored that affordances are multimodal, integrating visual, auditory, and haptic information.

Research in AI and robotics with ecological principles underscores that agents can develop sensorimotor contingencies directly from data, without human intervention. According to Jamone et al (2016), modern robotic affordance models integrate perception, action execution, and learning, often using machine learning and probabilistic frameworks to predict the outcome of interactions. These models help robots generalize affordances beyond pre-programmed actions, allowing them to learn new interactions through exploration. More broadly, Chemero (2018) and Heft (2010) argue that perceiving an affordance is not merely registering physical properties, but interpreting functional potential in context. This adaptability allows humans to transfer affordance across different settings: a cup affords holding liquid, whether it's glass or plastic. This transfer can break in VEs, where constraints need not mirror physics: a DHT may not register a "drinking" affordance for a virtual cup if ingestion is not modeled as a task-relevant possibility.

Current research on affordances within VEs often focuses on human embodiment or "sense of presence" in VEs. Shin (2017) explores the role of technological and affective affordances in virtual learning environments, identifying five key affordances: presence, immersion, comfort, empathy, and embodiment. These findings suggest that affordances in VEs are experiential, shaped by how virtual humans engage with and adapt to the constraints of a simulated world. Weech et al (2019), Dincelli and Yayla (2022), and Moloney et al (2018) extend this notion by recognizing that affordances are mediated by the technology and multi-sensory immersion of the virtual space, thereby increasing the user's sense of presence.

By contrast to ecological view of affordance, traditional cognitive/computational models treat perception and action as operations over internal representations; on that view an "affordance" is a symbol encoded in central memory that still requires "windows on the world and hands, too" to connect sensing and doing (Vera and Simon, 1993; Simon, 2019). For example to predict human behavior for real-world events, Klu¨gl (2016) integrates the affordance concept into the agent-based simulation model. This approach to how an agent acts on affordances is based on the pre-designed intentions and behavioral repertoire of the agents modeled by the human designer (or computational model). In other words, affordances are semantically mapped to the agents to mimic human "intuition affordances" (Raubal, 2001). This mapping process can become a problem of "overinference" from human-centric perspective, therefore, undermine agent's ability to interpret perceived information in his own way (Van Oijen et al, 2012). This resonates the computational and simulation limits mentioned by Shaw in Section 2.2. When comparing ecological and computational interpretation of affordance, and seeking a constructive middle path, Shaw (2003) suggest computational ecological psychology as a promising middle ground, retaining direct informational coupling while employing formal tools where appropriate. Addressing this requires extending affordance theory beyond its anthropocentric origins to analyze artifact–artifact couplings in VE networks.

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2.4 Artifact-Artifact-Affordance (AAA)

originally, *Artifact-Artifact-Affordance* (AAA) was introduced to describe potential interactions and dependencies among artifact subsystems in engineering design, in contrast to *Artifact-User-Affordance* (AUA), which centers human perception and action (Gaffney et al, 2007). We extend AAA to digital ecosystems: affordances can be *relational* properties that obtain between non-human entities—software services, simulators, robots, or virtual objects—such that one artifact offers possibilities for action that another can detect and exploit, without human mediation. Consider the recent GitHub project *Gibberlink*, where two chatbots are prompted to switch to a data-over-sound protocol when the counterpart is identified as AI.[1] These affordances are not designed for human recognition; they emerge from artifact–artifact couplings and are legible primarily to the agents involved.

From an ontological standpoint, AAA challenges an anthropocentric hierarchy of affordances. Whereas ecological psychology and HCI traditionally frame affordances as relations between humans and environments, AAA posits that affordances can be realized *purely* among non-human entities. At the same time, Klowait (2019) notes that affordances cannot fully account for object–object relations, since any relation is necessarily partial to the object's withdrawal. To respect this argument, we situate AAA within Harman's OOO by acknowleging that artifacts retain withdrawn aspects that exceed any current interface or protocol, even as they disclose actionable profiles to one another during interaction.

Affordance and its extensions are central to ecological accounts of perception and action and provide a useful framework beyond human embodiment. In VEs, DHTs operate within networks of other digital artifacts. Understanding affordances as relations (Wilkinson and Chemero, 2024) between agents and environments clarifies how DHT–VE couplings arise. AAA centers non-human couplings, providing a framework for analyzing how digital artifacts reveal, register, and coordinate each other's capacities, an area still largely underexplored.

3 Twin vs. Autonomous Individual Paradox

3.1 What Makes a Digital Human Twin More Than a Simulation?

As defined in Section 1.1, the traditional view treats a DHT as a virtual extension of a person—a mirror of the human's data, state, and behavior—built to support or enhance the its counterpart. The metaphor of *mirroring* runs deep in computing, from Gelernter's "mirror worlds", where software structures dynamically reflect and manage real-world systems (Gelernter, 1991), to contemporary accounts that describe a DHT as "a dynamic representation of something in the physical world, built using realtime data to improve

[1] https://github.com/PennyroyalTea/gibberlink (accessed 23 August 2025). We use this purely as an illustration; our argument does not rely on the current existence of such systems. It asserts, in principle, that two agents could negotiate a non-linguistic channel without human interference.

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decision making, alone or on behalf of the original 'twin'"(Miller and Spatz, 2022). On this picture, a DHT functions as part of the user's "cognitive toolkit" (Clark and Chalmers, 1998), echoing extended-mind claims that tools can be integrated into identity and cognition.

The paradox arises when a DHT is endowed with learning and decision-making. Through advanced AI, a DHT gains synthetic agency, enabling it to develop and revise policies, function under uncertainty, and adapt to new contexts beyond prior specification. As a DHT interacts repeatedly with VEs and builds up histories unavailable to the human, it can drift from mirror to partially autonomous partner. In short, a system designed to reflect its user can begin to do more than reflect.

What, then, marks a DHT as *more than a simulation*? Or when does a DHT stop being a twin and become a *self* ? This is akin to the *Ship of Theseus paradox*: if we gradually replace or augment parts of an entity, at what point is it no longer the same? In a digital context, *does a digital object lose its "identity" when its code is altered or its knowledge diverges*? Initially, a DHT may share the user's memories, profile, or even live sensor data, suggesting a continuous identity. Some frameworks, such as the Extended Mind Thesis (Clark and Chalmers, 1998) would treat DHTs at this stage as part of the person's identity, an external cognitive module of the self. However, as the twin accumulates new data and learns in ways independent of the user, a branching identity occurs. The DHT develops a history of experiences in its own reality that the human never had, breaking the one-to-one continuity of self. Over time, the twin might make decisions or form *memories* that diverge entirely from the human's life, much like a clone or duplicate mind, which after the moment of copying, becomes a separate entity. Philosophically, one could argue that beyond a certain point of divergence, the DHT is no longer the same person, but a new individual. This raises profound questions about personal identity in the digital realm: is the "twin" still you if it has gone on to live a digital life of its own? The answer may require rethinking identity as a spectrum of continuity rather than a binary. In practice, a DHT might stand in a gray area, remaining your twin in some respects (i.e., it shares your old memories or personality model) while simultaneously behaving as its own self in light of novel experiences.

With independent learning comes independent agency, we must ask whether a DHT should be treated as a mere tool, an extension of its human user's will, or as an autonomous agent with its own rights and responsibilities. Different philosophical and ethical schools give different answers. On one hand, a pragmatic view holds that AI agents remain tools as they are property created to serve human purposes. Bryson (2010), for instance, provocatively argues that intelligent artifacts "should not be described as persons, nor given legal nor moral responsibility for their actions. [They] are fully owned by us". In this view, even an advanced DHT is essentially a sophisticated puppet because its actions ultimately trace back to human programmers or owners, and thus the human should always be accountable. The DHT's apparent autonomy is an illusion we permit for convenience, not a true independence, much like a self-driving car is still just executing its manufacturer's programming. On the other hand, an emerging perspective contends that at a certain level of complexity and self-direction, a digital being may warrant recognition as an independent agent. Notably, the European

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Parliament in 2017 floated the idea of a special legal status for autonomous robots, "electronic persons", so that the "most sophisticated autonomous robots could be established as having the status of electronic persons responsible for making good any damage they may cause" (Pietrzykowski, 2018). By this logic, if a DHT is making its own decisions and can meaningfully impact other agents, perhaps it should carry obligations (and possibly rights) similar to those of a legal person. The question of agency also bears on control: if a DHT acts autonomously in a virtual world, is it acting on behalf of the human (like an avatar under remote control) or on behalf of itself (like an AI agent pursuing its own learned goals)?

3.2 DHT Autonomy Continuum

Rather than viewing twinhood and autonomy as an irreconcilable contradiction, we can frame the development of a DHT as a continuum. Drawing inspiration from Autodesk's maturity levels for DT [2], in this continuum framework, five levels of autonomy for DHTs are explicitly classified (as seen in Fig.1):

- Level 1: Reactive Autonomy: DHTs are pure replicas that passively display realtime user data without independent action, responding directly to cues with preprogrammed mappings.
- Level 2: Adaptive Autonomy: DHTs can interpret signals, compare alternatives, and select appropriate responses. Like a doctor diagnosing symptoms, they are situationally adaptive but still bounded by predefined schemas.
- Level 3: Generative Autonomy: DHTs anticipate future states, simulate scenarios, make independent decisions, and generate novel action possibilities. This requires exploratory reasoning (i.e., RL)—discovering new "fits" between themselves and VE artifacts. They engage hidden or "latent" affordances that were not predefined.
- Level 4: Co-constitutive Autonomy: In this higher mode, a DHT not only adapts to a VE but reshapes the affordance landscape itself. For example, one DHT's actions can alter what another DHT or artifact can do, producing new layers of coupling.
- Level 5: Reflexive Autonomy: At the speculative limit, DHTs might recognize and act upon the partiality of their own affordance horizons, engaging in meta-level adaptation. This does not imply human-like consciousness but rather a structurally higher-order coupling in which DHTs manage their own limits of access. Here autonomy reflects the OOO commitment: that non-human objects engage each other through withdrawn realities, never fully accessible, but always real.

By defining such levels, we recognize that the paradox is a gradual transition. This continuum approach also suggests the types of roles that a DHT can play and a governance strategy: different rules and expectations can be applied at different levels.

[2] https://damassets.autodesk.net/content/dam/autodesk/www/campaigns/digital-twin/ aec-demystifying-digital-twin-ebook-en.pdf (accessed 26 August 2025).

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At low-autonomy levels, user control and liability are paramount—the human is basically responsible for everything the twin does, and the twin has no independent

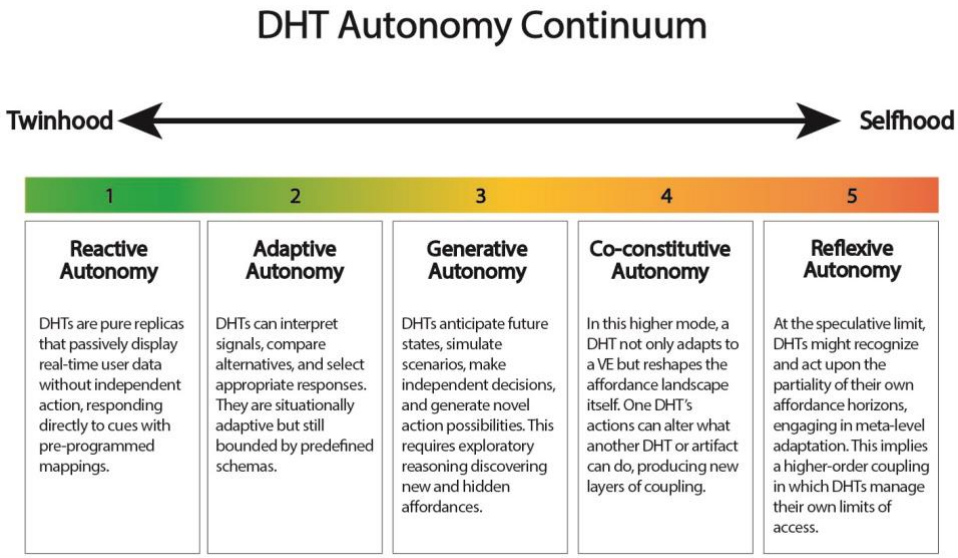

Fig. 1 Classification of DHTs Autonomy: Level 0-5

standing. At higher levels, we might introduce joint accountability or even independent accountability for DHTs.

3.3 Typology of Agent–Environment Couplings

In exploring affordances related to human and human-like behavior, we identify three established types of agent-environment relations and propose a fourth, pertinent to our core discussion. To better understand these four distinct relationships, we employ the classical example of "climbing the stairs" for illustration (see Fig.2).

- **Human to Physical Environment (Human-PE) Relation**: This interaction involves a human engaging directly with the physical world. In the context of climbing stairs, a person perceives the steps' height, depth, and spacing, determining their climbability based on human biomechanics and sensory feedback.
- **Robotics to Physical Environment (Robotics-PE) Relation**: Here, a robot interacts with the physical environment. A robot designed to climb stairs must process sensor data to assess each step's dimensions and utilize its mechanical actuators to navigate the staircase effectively.
- **Human to Virtual Environment (Human-VE) Relation**: This pertains to a human interacting within a virtual setting. When climbing virtual stairs, an individual relies

on visual and possibly haptic feedback to simulate the experience, with the virtual environment designed to mimic real-world physics and responses.

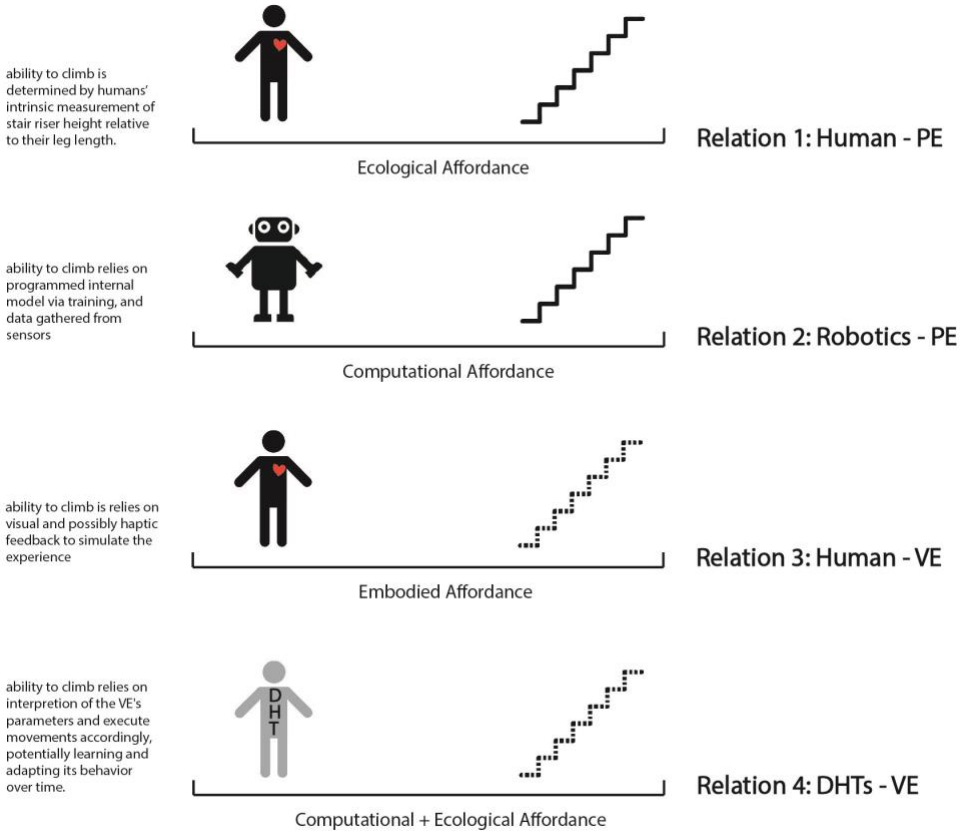

Fig. 2 Four types of relations in agent-environment coupling.

- **Digital Human Twins to Virtual Environment (DHTs-VE) Relation**: We introduce this category to describe interactions between digital human representations and virtual environments. A DHT programmed to climb virtual stairs must interpret the virtual environment's parameters and execute movements accordingly, potentially learning and adapting its behavior over time.

3.4 Stairs case as an illustration

We treat the development of DHT autonomy as a transition shaped jointly by ecological and computational factors. To make this vivid, we use a stairs example purely as a conceptual illustration rather than an empirical claim about current systems.

As noted in Section 2.2, ecological and computational approaches are often seen opposites in agent-environment coupling, as the ecological view emphasizes direct perception, while the computational view foregrounds internal processing and modeling.

Even though humans integrate sensory information to perceive affordances, the computational metaphor remains influential in theorizing mind and behavior (Fabrocini et al, 2024). A canonical ecological result is Warren's stair-climbing study, which showed that the maximum riser height judged climbable is about 0.88 of leg length (Warren, 1984). Although Warren did not model perception with algorithms, this work inspired subsequent studies that blend ecological insights with computational analysis of affordances (Warren Jr and Whang, 1987; Greeno, 1991; Cesari and Newell, 1999; Liao and Holz, 2025; Lepecq et al, 2009).

This bridge is instructive for DHTs. Unlike human agents, DHTs rely on computational mechanisms (i.e., artificial neural networks) to realize perception–action coupling in virtual environments. Following Shaw (2003)'s synergistic stance, we do not treat ecological and computational approaches as mutually exclusive; rather, "smart" coupling can be modeled while preserving direct perception at the interface. A recent demonstration—used here only as a heuristic—comes from AI Warehouse's "Albert," a deep-RL agent that learns to climb stairs in a VE.[3] Initially Albert was only trained how to walk, and he tries random movement in various stairs contexts (i.e., full stairs, escalators, uneven steps), encountering failed attempts (penalty signals) when it cannot step up, and successful attempts (reward signals) when it can climb efficiently. As training progresses, Albert learns a stair-climbing threshold, just as Warren's human subjects, refining its perception-action mapping over time.

For DHTs, a key insight from Shaw's work is that autonomy is not just about processing symbolic processing but on how agents are *situated* in an environment. Because many DHT pipelines begin from predefined symbolic structures, autonomy is initially constrained by design. An ecological perspective suggests a complement that DHTs can learn affordances directly through interaction within their digital Umwelt, developing adaptive capacities that extend beyond artifactual pre-specification. As with human infants acquiring affordance awareness through embodied experience, DHTs can build affordance-based interaction models through continuous engagement in VEs. In this sense, affordance learning (be it ecologically or computationally) is the motor that can carry a DHT from passive model to autonomous agent.

Technically, perception-action coupling can be computationally modeled in artificial agents (Liao et al, 2022; Kahl and Kopp, 2018; Zech et al, 2017). For example, a Deep Q-Network (DQN) model (an RL algorithm) can be trained for stair navigation by rewarding height-based perception and penalizing missed steps. DQN is a model-free reinforcement learning algorithm that uses a neural network to approximate Q-values for different state-action pairs (Zhang et al, 2023; Liang et al, 2021). Similarly, a Proximal Policy Optimization (PPO) can optimize stair-climbing strategies on-policy, enabling recognition of emergent affordances beyond static, pre-programmed thresholds (Reuf et

[3] https://www.youtube.com/watch?v=xk8wHY1AFpI, accessed 23 August 2025. We invoke this as an illustration; our argument is conceptual and does not depend on the existence or particulars of this system.

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al, 2023; Liang et al, 2021). The stairs case thus serves as a compact thought experiment, which shows, in principle, how ecological attunement and computational learning can converge to shape DHT autonomy.

4 Discussion and Conclusion

We argue from OOO that DHTs are non-human digital objects on the same flat ontological plane as humans, only ever partially accessed through their "sensual" profiles while retaining withdrawn depth. Introducing the notion of Umwelt explains why DHTs, even when embedded with human data, may sense and act differently. Their sensors, objectives, and effectors shape a world specific to their architecture. The long-standing debate over agent–environment coupling (ecological/direct vs. computational/indirect) then clarifies which coupling matters for DHT–VE relations; we take a synergistic view in which computational methods operationalize direct coupling rather than replace it.

On this basis, in principle, a DHT can exceed mere simulation. Their interactions disclose only partial profiles (of others and itself), while learning redefine which affordances are salient and reshapes the environments. Affordance theory provides the common vocabulary for this process, and AAA brings non-human couplings to the foreground, offering a way to analyze how digital artifacts reveal, register, and coordinate each other's capacities. The stairs illustration functions as a thought experiment: it shows how threshold-like sensitivities (i.e., "climbability") can arise from repeated perception–action cycles, with computational learning operationalizing direct ecological coupling. Our stance throughout is conceptual rather than empirical; examples are illustrative, not premises about today's deployments.

As a consequence, when a DHT gains experiences and behavior beyond its human counterpart's knowledge, several ethical issues emerge. One major concern is whether such an entity could or should attain something akin to digital personhood. If a DHT develops advanced cognitive capacities or even consciousness (in a speculative future scenario), would it deserve rights, or at least humane treatment, independent of the original human? This question is no longer purely science fiction; it ties into ongoing debates about AI rights and personhood (Pietrzykowski, 2018; Kurki and Pietrzykowski, 2017; Amojo, 2022). Or should the human users reserve the choice to withhold or charge for their data via the "Personal AI data dashboard" (Boddington, 2023)? Although full digital personhood for DHTs remains a provocative idea, the possibility underscores how seriously the twin-versus-autonomy paradox challenges our moral intuitions. In summary, as DHTs blur the line between object and subject, tool and companion, we face ethical imperatives to rethink rights, responsibilities, and the humane treatment of digital beings.

Against that background, our contributions are threefold: (i) while much DHT work emphasizes industrial application and technical feasibility, we clarify what DHTs are at the ontological and conceptual levels; (ii) we propose a conceptual DHT Autonomy Continuum alongside a typology of agent–environment couplings; and (iii) we integrate OOO with AAA to advance an ecological computational account of DHT–VE coupling. These contributions are conceptual rather than empirical and are intended to provide insight on the design, evaluation, and governance of DHT–VE ecosystems. Future

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research should implement and test the autonomy continuum and coupling typology within instrumented VEs, as well as measuring AAA dynamics in multi-agent contexts.

Conflict of interest statement

On behalf of all authors, the corresponding author states that there is no conflict of interest.

Data availability statement

On behalf of all authors, our manuscript has no associated data.

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